DBA5103

Operations Research and Analytics

Zhenyu Hu

Semester I, 2026/2027



Software Implementation

Wyndor Glass Co.

TABLE 3.1 Data for the Wyndor Glass Co. problem

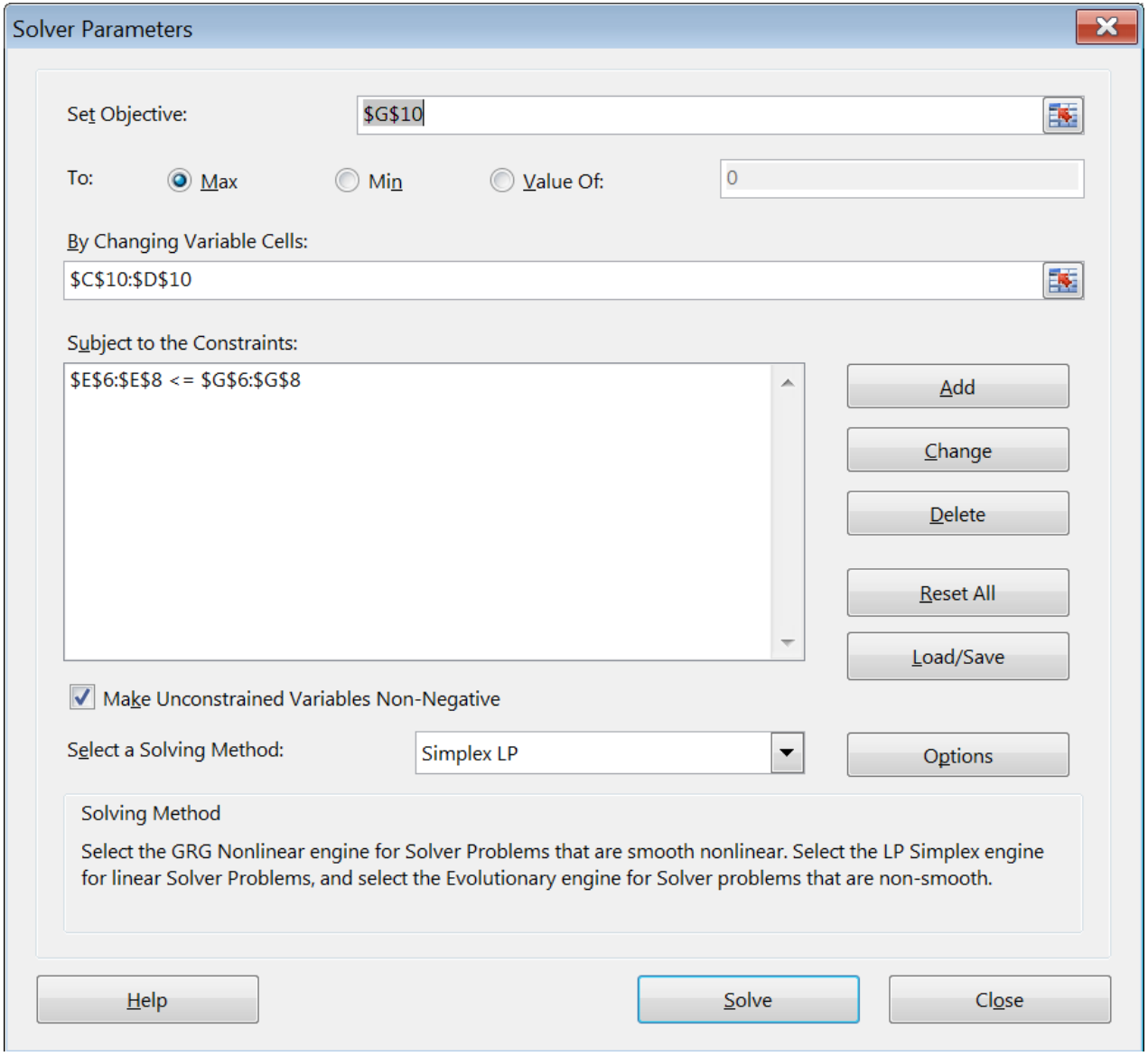
Plant	Production Time per Batch, Hours: Product 1	Production Time per Batch, Hours: Product 2	Production Time Available per Week, Hours
1	1	0	4
2	0	2	12
3	3	2	18
Profit per batch	$3,000	$5,000	

$$max\ 3x_1 + 5x_2$$
$$x_1 \leq 4,$$
$$2x_2 \leq 12,$$
$$3x_1 + 2x_2 \leq 18,$$
$$x_1 \geq 0, x_2 \geq 0$$

Model in Solver

	A	B	C	D	E	F	G	H
1			Doors	Windows				
2		Profit per batch ($1,000)	3	5				
3								
4								
5			Hours Used per Batch Produced		Hours Used		Hours Available	
6		Plant 1	1	0	0	<=	4	
7		Plant 2	0	2	0	<=	12	
8		Plant 3	3	2	0	<=	18	
9								
10		Batches Produced	0	0		Total Profit	0	
11								

Model in Solver



Model in Gurobi

```
from gurobipy import *

m = Model("Wyndor")

# Creat variables
# addVar(lb=0.0, ub=GRB.INFINITY, obj=0.0, vtype=GRB.CONTINUOUS, name="", column=None)
x1 = m.addVar(name = "x1")
x2 = m.addVar(name = "x2")

# Set objective
m.setObjective(3*x1 + 5*x2, GRB.MAXIMIZE)

# Add constraint:
m.addConstr(x1 <= 4, "Plant1")

m.addConstr(2*x2 <= 12, "Plant2")

m.addConstr(3*x1 + 2*x2 <= 18, "Plant3")
```

Solving the Model and Display Output

- **Solving the model**

```
# Solving the model
m.optimize()
```

- **Display solution and optimal value**

```
# Print optimal solutions and optimal value
print(x1.VarName, x1.x)
print(x2.VarName, x2.x)

print('Obj:', m.objVal)
```

Separating Model and Data: Model

```
#########Model Set-up###############

m = Model("Wyndor")

# Creat variables
# addVars ( *indices, lb=0.0, ub=GRB.INFINITY, obj=0.0, vtype=GRB.CONTINUOUS, name="" )

x = m.addVars(N, name = "x")

# Set objective
m.setObjective( quicksum(profit[i]*x[i] for i in range(N)), GRB.MAXIMIZE)

# Add constraints:
m.addConstrs(( quicksum(rate[i,j]*x[j] for j in range(N)) <= capacity[i] for i in range(M) ), "Plant")
```

Separating Model and Data: Data

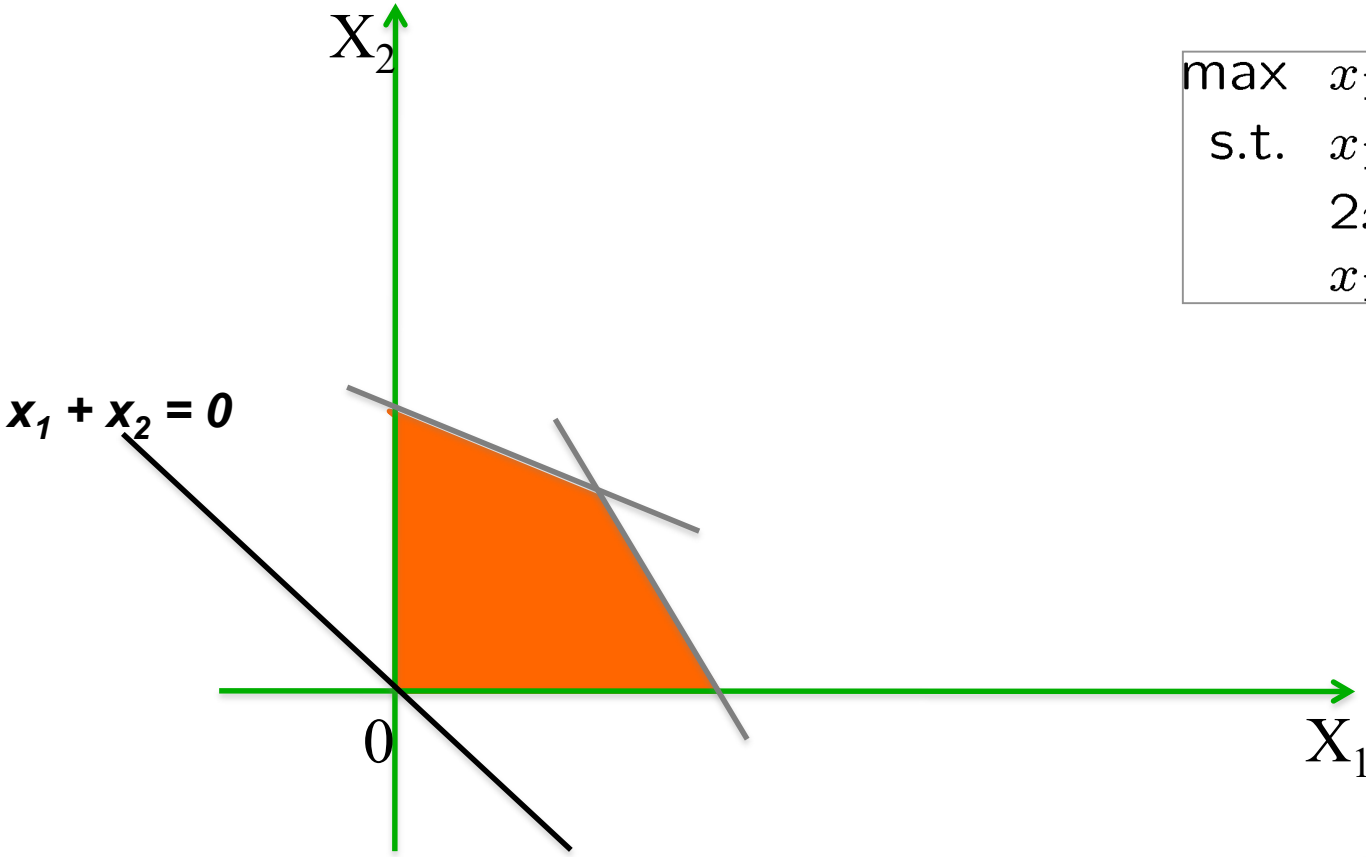
```
# Objective coefficient: profit for each product
profit = np.array([3, 5])
# Constraint right-hand-side: capacity for each plant
capacity = np.array([4, 12, 18])

# A matrix: the consumption of capacity at plant i of product j
rate = np.array([[1, 0],
                 [0, 2],
                 [3, 2]])
# From A matrix, extract the number of products: N and the number of plants: M
M, N = rate.shape
```

Geometry of Linear Programming

Geometry of LP in 2D

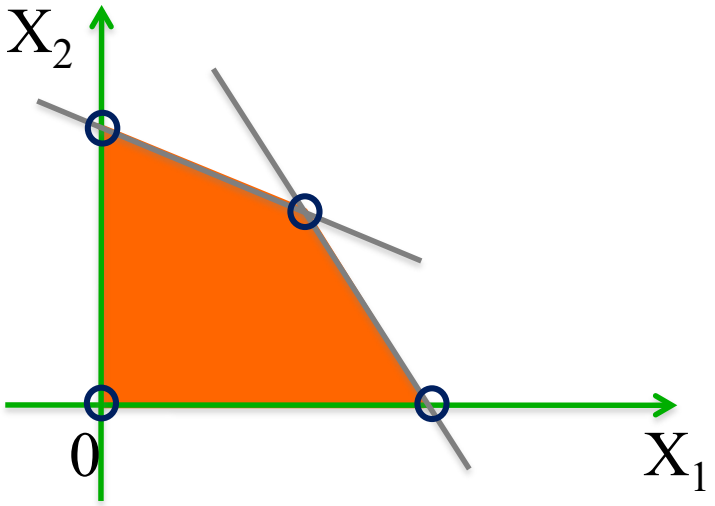
- **Objective: *maximize $x_1 + x_2$***



$$\begin{aligned} \max \quad & x_1 + x_2 \\ \text{s.t.} \quad & x_1 + 2x_2 \leq 3 \\ & 2x_1 + x_2 \leq 3 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Geometry of LP in 2D

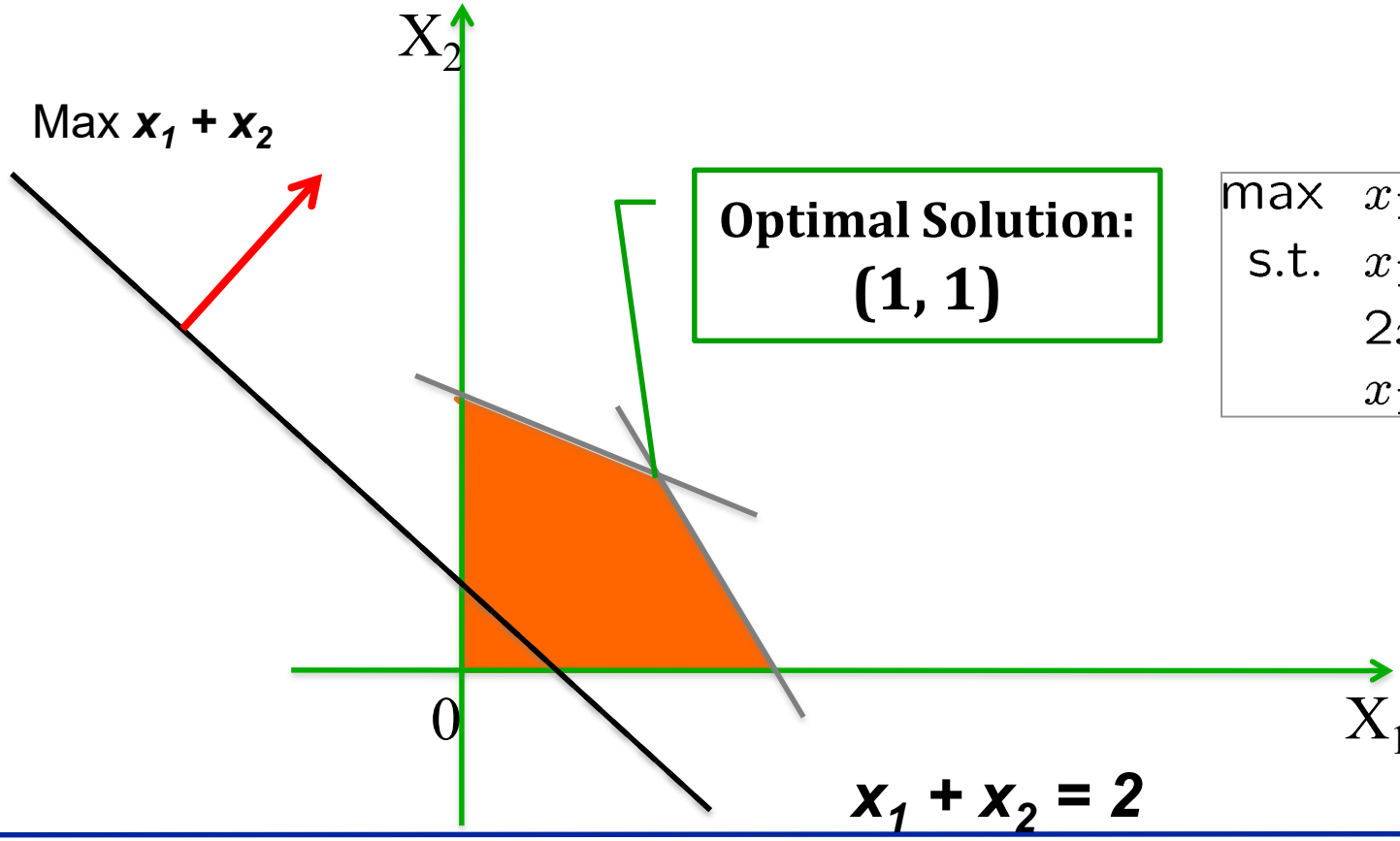
- **Feasible Region: The set of all the allowed solutions; a polyhedron described by the constraints**
  - **Each equality constraint defines a line**
  - **Each inequality constraint defines a half-space**
- **Extreme Points (Corner-Point Feasible Solution): Corner points on the boundary of the feasible region. E.g., (0, 0), (1.5, 0), (0, 1.5), and (1, 1)**
- **Infeasible problem: A problem with an empty feasible region**



Geometry of LP in 2D

- **Optimal Solution:** The ***best feasible*** **solution**

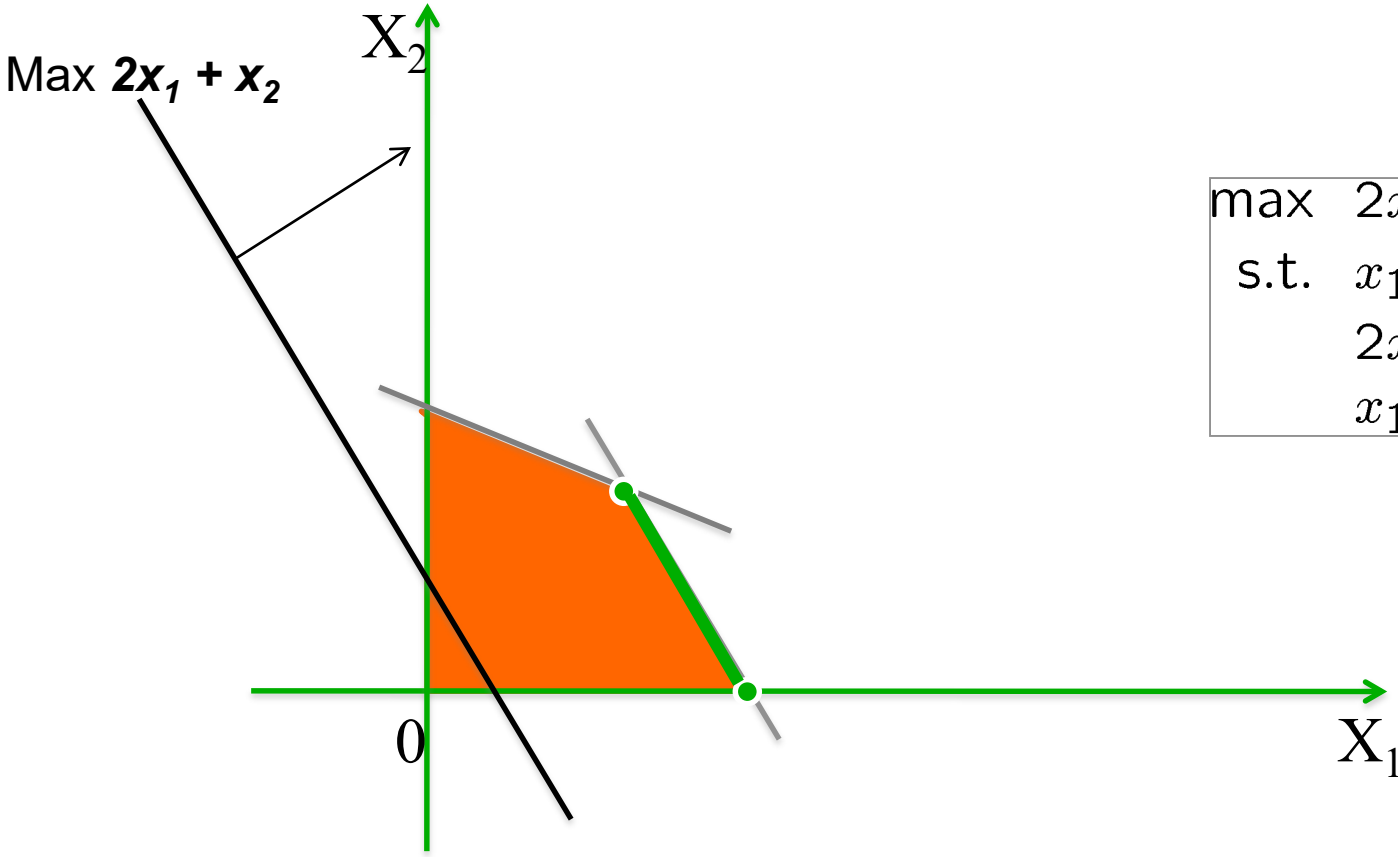
For any feasible LP with a finite optimal solution, there exists an optimal solution that is an extreme point



$$\begin{aligned} \max \quad & x_1 + x_2 \\ \text{s.t.} \quad & x_1 + 2x_2 \leq 3 \\ & 2x_1 + x_2 \leq 3 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Geometry of LP in 2D

- **Optimal solutions may NOT be <u>unique</u>**

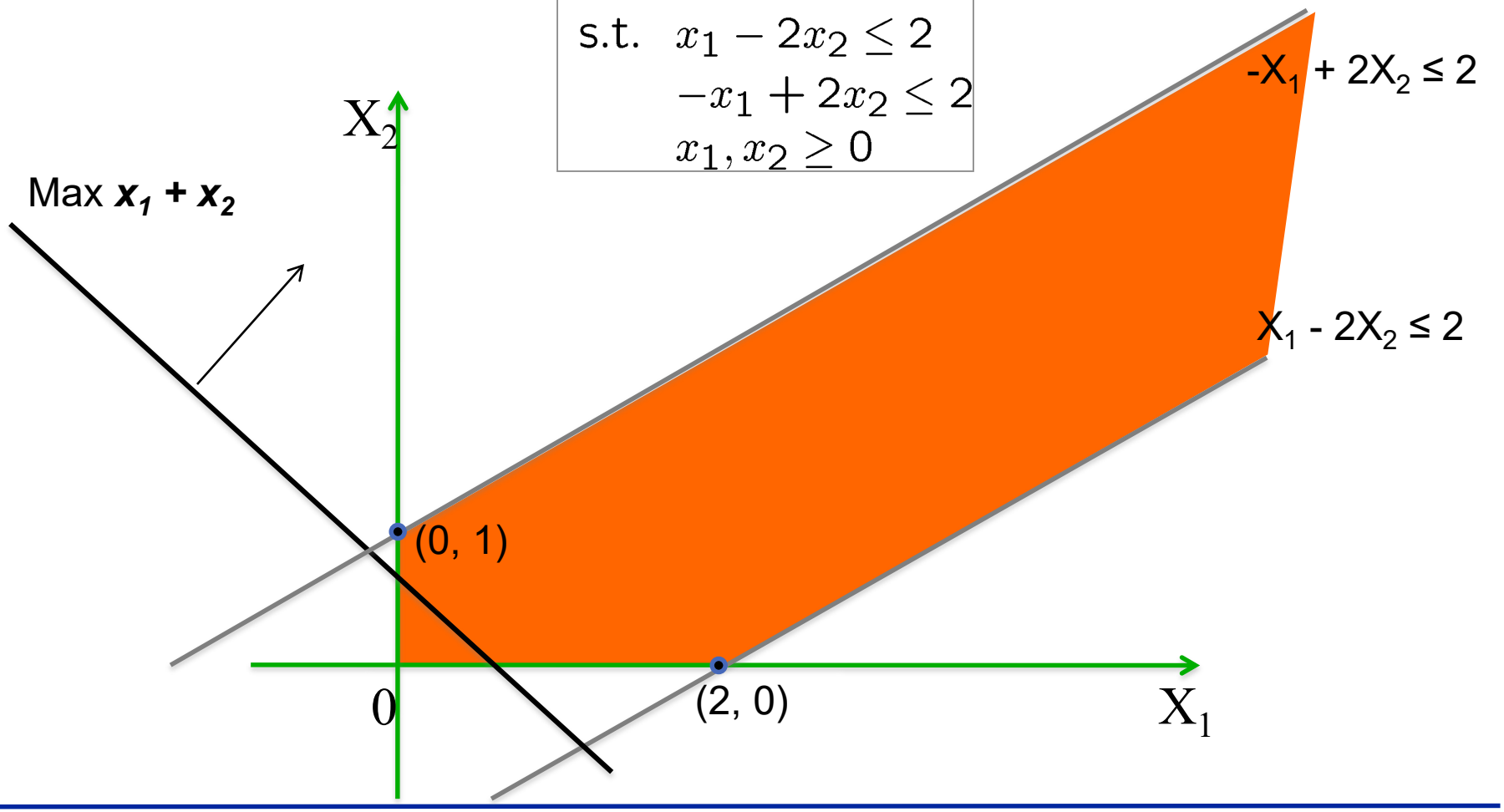


$$\begin{aligned} \max \quad & 2x_1 + x_2 \\ \text{s.t.} \quad & x_1 + 2x_2 \leq 3 \\ & 2x_1 + x_2 \leq 3 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Geometry of LP in 2D

- **Optimal solution may NOT be finite**

$$\begin{aligned} \max \quad & x_1 + x_2 \\ \text{s.t.} \quad & x_1 - 2x_2 \le 2 \\ & -x_1 + 2x_2 \le 2 \\ & x_1, x_2 \ge 0 \end{aligned}$$



Summary

- **Solutions of LP**
  - Unique optimal solution
  - Multiple optimal solutions
  - The optimal objective is not bounded
  - Infeasible

- **Sufficient to focus on extreme point**
  - How to describe extreme point in high dimension?

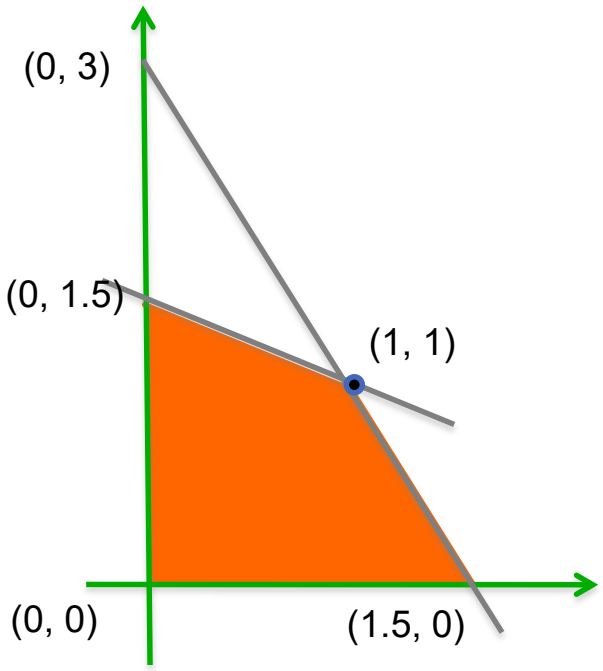
Active/Binding Constraints

- **Binding (or active) constraints: The constraints that are satisfied at *equality* for a given solution**
  - **All equality constraints (if satisfied) are binding**

- **Non-binding (or inactive) constraints are satisfied at *strict inequality* for a given solution**

- **The inequality level is known as the slack**
  - **Binding constraints have zero slack by definition**

Back to Extreme Point



$$\begin{aligned} \max \quad & x_1 + x_2 \\ \text{s.t.} \quad & x_1 + 2x_2 \leq 3 \\ & 2x_1 + x_2 \leq 3 \\ & x_1, x_2 \geq 0 \end{aligned}$$

Constraints	Binding?	Slack
$X_1 + 2X_2 \leq 3$		
$2X_1 + X_2 \leq 3$		
$X_1 \geq 0$		
$X_2 \geq 0$		

How many binding constraints will an extreme point have?

What if we add constraints $x_2 \leq 1.5$ and $2x_1 \geq 0$? Think about (0,1).

What about (0,3)?

Extreme Point: An Algebraic Perspective

Consider a feasible region in n-dimensional space:

Polyhedron:

$$a_{i1}x_1 + \cdots + a_{in}x_n \geq b_i, i \in M_1$$

$$a_{j1}x_1 + \cdots + a_{jn}x_n \leq b_j, j \in M_2$$

$$a_{k1}x_1 + \cdots + a_{kn}x_n = b_k, k \in M_3$$

- **A vector $x^* = (x_1^*, \ldots, x_n^*)$ is called basic solution if**
  - **All equality constraints are active;**
  - **Out of all constraints that are active at x^*, there are n of them that are linearly independent.**
- **x^* is called a basic feasible solution (BFS) if it is a basic solution and it is feasible.**

BFS = Extreme Point

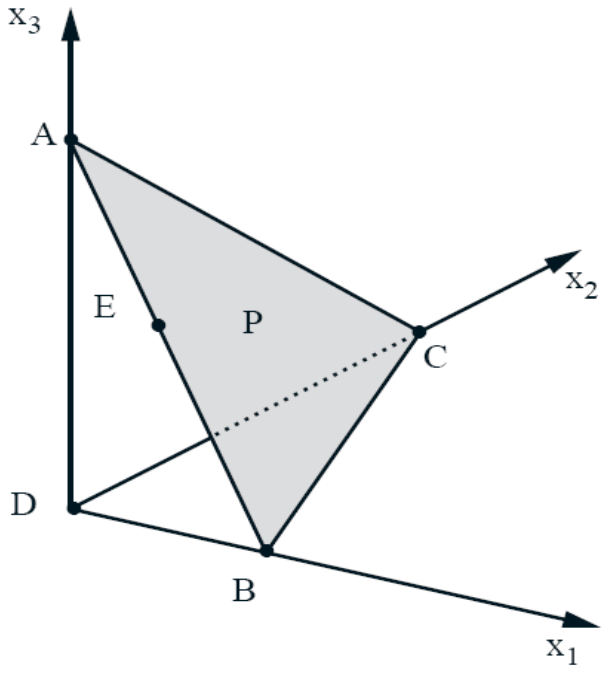
3-D Example

Consider a feasible region in 3-dimensional space:

$$x_1 + x_2 + x_3 = 1,$$

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$$

- **Which points are basic solutions, basic feasible solutions?**



3-D Example

Consider a feasible region in 3-dimensional space

$$x_1 \leq 1, x_2 \leq 1, x_3 \leq 1,$$

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$$

- **What are the extreme points/BFS?**

Insights: Portfolio Management Problem

- There are 100 stocks: stock i has return r_i per dollar invested.
- Budget constraint: b dollars available.
- How many stocks to invest and how to budget?

$$\begin{aligned} max \quad & r_1 x_1 + \cdots + r_{100} x_{100} \\ s.t. \quad & x_1 + \cdots + x_{100} \leq b, \\ & x_1 \geq 0, \ldots, x_{100} \geq 0 \end{aligned}$$

Insights: Portfolio Management Problem

- Suppose the stocks 1-10 are the top 10% riskiest stocks and at most half of the budget can be allocated to these stocks.
- How many stocks to invest and how to budget?

$$
\begin{aligned}
max \quad & r_1 x_1 + \cdots + r_{100} x_{100} \\
s.t. \quad & x_1 + \cdots + x_{100} \leq b, \\
& x_1 + \cdots + x_{10} \leq b/2, \\
& x_1 \geq 0, \ldots, x_{100} \geq 0
\end{aligned}
$$

Extreme Point in Standard Form

- For general form, it is not easy to check whether a basic solution is feasible or not; so consider

$$Ax = b$$

$$x \geq 0$$

- A is an $m \times n$ matrix. Assume $m \leq n$ and the m rows of A are linearly independent.
- A vector x is a basic solution if $Ax = b$ and there exist indices $B(1), \ldots, B(m)$ such that
  - The columns $A_{B(1)}, \ldots, A_{B(m)}$ are linearly independent;
  - If $i \neq B(1), \ldots, B(m)$, then $x_i = 0$.

Some Terminology

- The columns $A_{B(1)}, \ldots, A_{B(m)}$ are called **basic columns**.
- The $m \times m$ matrix

$$B = [A_{B(1)}, \ldots, A_{B(m)}],$$

  is called the **basis matrix**.
- The corresponding components of the vector x: $x_B = (x_{B(1)}, \ldots, x_{B(m)})$ are called **basic variables**; the remaining components are called **nonbasic**.
  - $x_B = B^{-1}b$
  - If x_i is a nonbasic variable, then $x_i = 0$.

An algorithm to find extreme point

1. Convert an LP to standard form

$$Ax = b$$
$$x \geq 0$$

2. Choose m linearly independent columns $A_{B(1)}, \ldots, A_{B(m)}$ from the matrix A.
3. Let $x_i = 0$ if $i \neq B(1), \ldots, B(m)$.
4. Solve the system of m equations $Ax = b$ for the unknowns $x_{B(1)}, \ldots, x_{B(m)}$.
5. If x_i are all non-negative ($x \geq 0$), then we have found an extreme point (BFS); otherwise go back to step 2.

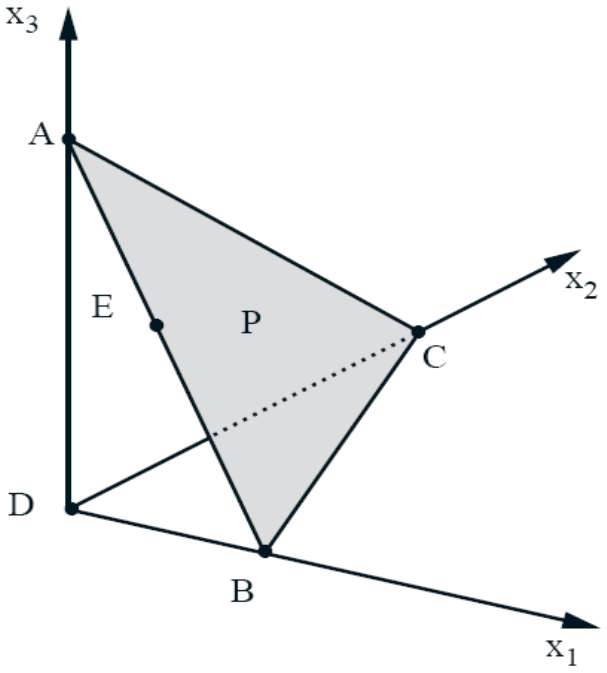
Example

Consider following standard form:

$$x_1 + x_2 + x_3 = 1,$$

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$$

- **A is a 1×3 matrix (m=1):**
  - **Choose column 1;**
  - **Let $x_2 = 0, x_3 = 0$;**
  - **Solve $x_1 = 1$;**
  - **Check non-negativity.**



Example

$$\begin{bmatrix} 1 & 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 6 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \boldsymbol{x} = \begin{bmatrix} 8 \\ 12 \\ 4 \\ 6 \end{bmatrix}, \boldsymbol{x} \geq \mathbf{0}.$$

- **How to find a BFS?**
- **Can the constraints be expressed in inequalities?**

Summary

- Each basic solution has *m* basic variables
  - Rest of variables are nonbasic variables and are set to zero
- Values of basic variables given by:
  - Solution to a system of *m* equations
- A basic feasible solution (BFS)
  - Basic solution where all *m* basic variables are nonnegative

Simplex Method

Naïve Approach

- **Consider an LP in standard form**

$$\begin{aligned} min \quad & c'x \\ s.t. \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

  1. **Find all extreme points.**
  2. **Compute $c'x$ at these extreme points.**
  3. **Pick the smallest one.**

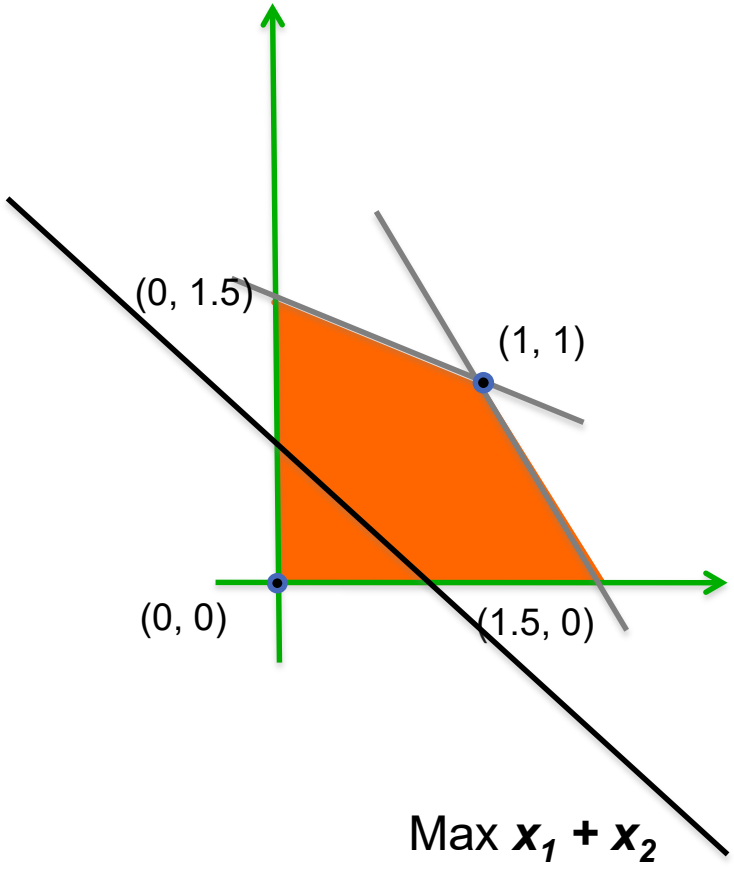
- **Reduces from infinite to finite; how many possible basic solutions are there?**
  - **n = 20, m = 10: about 0.2 million;**
  - **n = 30, m = 15: about 155 million.**

A General Idea

1. Find a feasible solution;
2. Determine whether the current solution is optimal or not: optimality condition.
   - If yes, terminate; else proceed to step 3.
3. Search the neighborhood of the current solution to find another feasible solution with lower cost, and go back to step 2.

Optimality Condition

- **Why is (0, 0) not optimal?**
- **Why is (1, 1) optimal?**



A General Optimization Problem

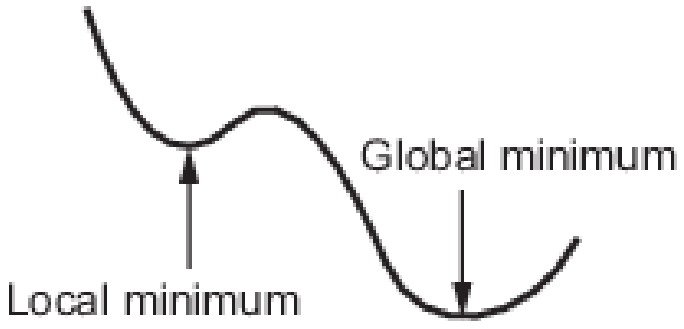
- **A general problem**

$$\begin{aligned} &min && f(x) \\ &s.t. && x \in \mathcal{F} \end{aligned}$$

- **For LP**
  - $f(x) = c'x$
  - $\mathcal{F} = \{x \,|Ax = b, x \geq 0\}$

Local Optimality vs Global Optimality

- For a feasible solution $x \in \mathcal{F}$, if no other feasible solution in the neighborhood of x leads to lower cost, x is called a local optimal solution.
- For a feasible solution $x \in \mathcal{F}$, if no other feasible solution in $\mathcal{F}$ leads to lower cost, x is called a global optimal solution.

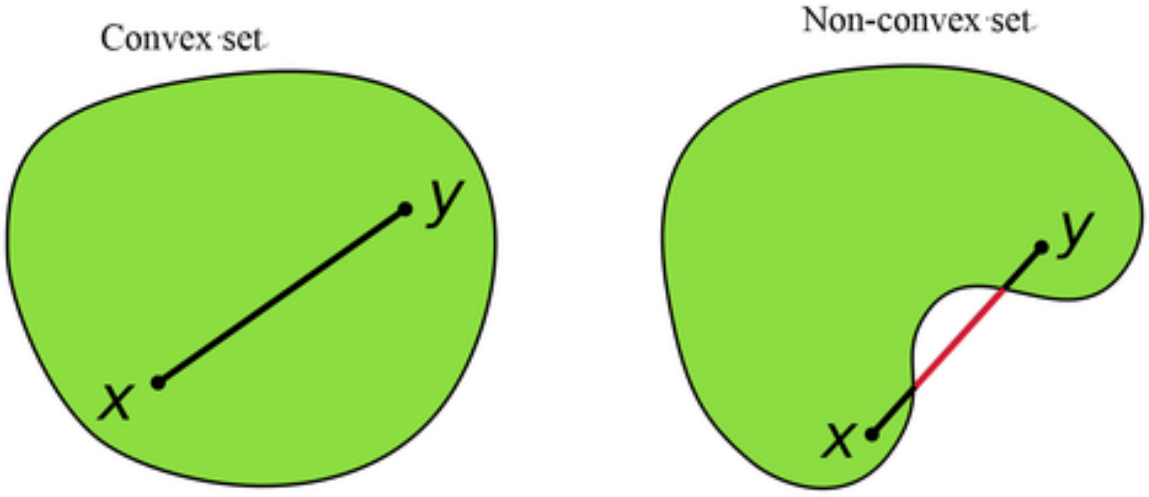


Convex Set and Convex Function

- **A set $\mathcal{F}$ is called convex set if for every $\lambda \in [0, 1]$ and $x, y \in \mathcal{F}$, we have**

$$\lambda x + (1 - \lambda) y \in \mathcal{F}$$

  - **Polyhedron $\{x \mid Ax = b, x \geq 0\}$ is a convex set.**

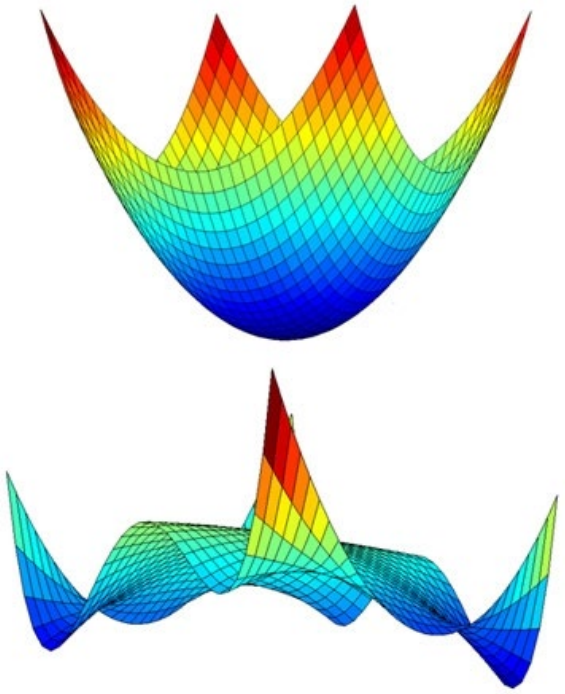


Convex Set and Convex Function

- A function $f(x)$ is called convex on a convex set $\mathcal{F}$ if for every $\lambda \in [0,1]$ and $x, y \in \mathcal{F}$, we have

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda) f(y)$$

  - A linear function $f(x) = c'x$ is a convex function.



A General Principle

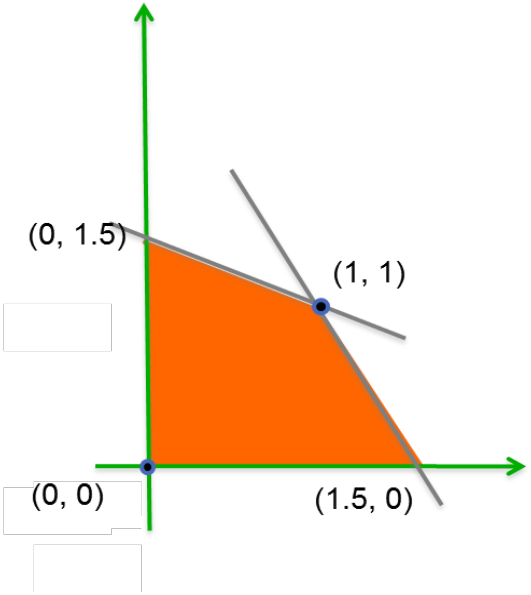
- If $\mathcal{F}$ is a convex set and $f(x)$ is a convex function, then

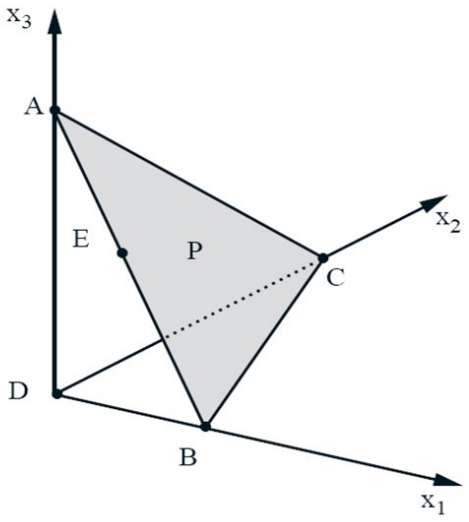
$$\begin{aligned} min \quad & f(x) \\ s.t. \quad & x \in \mathcal{F} \end{aligned}$$

  - is called a convex programming problem;
  - a local optimal solution is a global optimal solution.
- LP is a special case of convex programming.
  - It is sufficient to check feasible solution nearby.
  - It is sufficient to check BFS nearby.

BFS nearby: Adjacent BFS

- **Two distinct BFSs in n-dimensional space are said to be adjacent if they share n-1 linearly independent active constraints.**
  - **For standard form, this implies that all but one of their nonbasic variables are the same.**

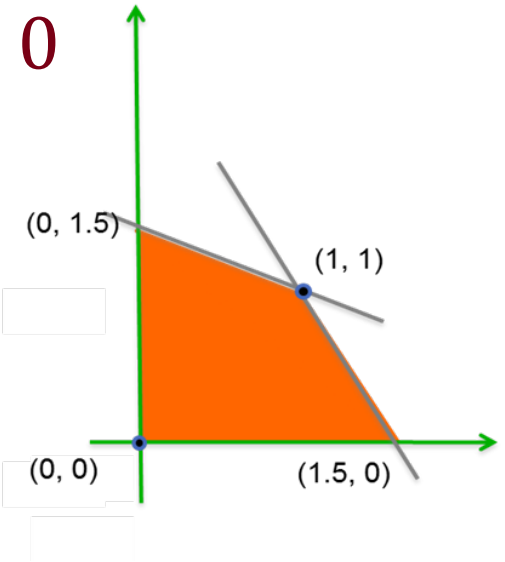




Example

$$x_1 + 2x_2 + x_3 = 3,$$
$$2x_1 + x_2 + x_4 = 3,$$
$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0$$

- **Which one is adjacent BFS**
  - $(0, 0, 3, 3), (0, 1.5, 0, 1.5)$
  - $(0, 0, 3, 3), (1, 1, 0, 0)$
  - $(1,1, 0, 0), (0, 1.5, 0, 1.5)$
  - $(1,1, 0, 0), (1.5, 0, 1.5, 0)$



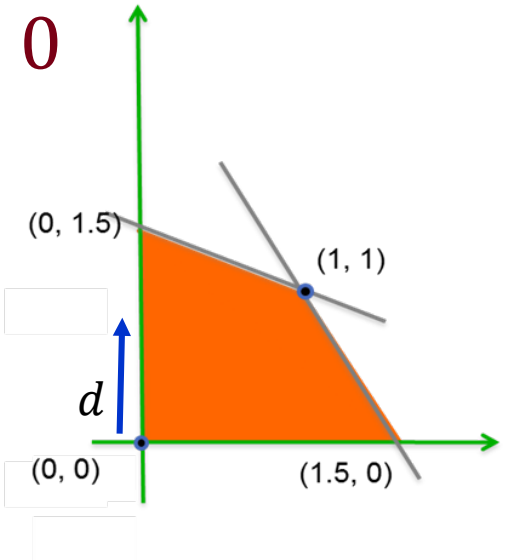
Example

$$\begin{bmatrix} 1 & 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 6 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \boldsymbol{x} = \begin{bmatrix} 8 \\ 12 \\ 4 \\ 6 \end{bmatrix}, \boldsymbol{x} \geq \mathbf{0}.$$

- **Which one is adjacent BFS**
  - $(0, 0, 0, 8, 12, 4, 6), (4, 0, 0, 4, 12, 0, 6)$
  - $(0, 0, 0, 8, 12, 4, 6), (0, 6, 0, 2, 6, 4, 0)$
  - $(0, 6, 0, 2, 6, 4, 0), (4, 0, 0, 4, 12, 0, 6)$

Example: Iteration 0

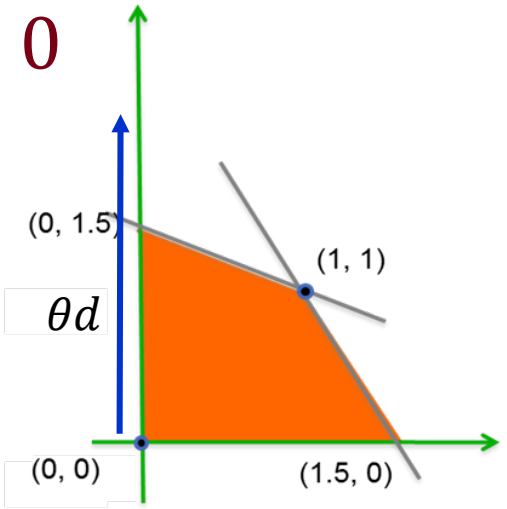
$$x_1 + 2x_2 + x_3 = 3,$$
$$2x_1 + x_2 + x_4 = 3,$$
$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0$$



- **Adjacent BFS**
  - $(0, 0, 3, 3) \rightarrow (0, ?, ?, ?)$
- **How to move?**
  - Stay on edge: $d = (0, 1, d_3, d_4)$
  - Stay feasible: $\begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{bmatrix} \left(\begin{bmatrix} 0 \\ 0 \\ 3 \\ 3 \end{bmatrix} + \theta \begin{bmatrix} 0 \\ 1 \\ d_3 \\ d_4 \end{bmatrix} \right) = \begin{bmatrix} 3 \\ 3 \end{bmatrix}, \theta \geq 0$
  - $\Rightarrow d = (0, 1, -2, -1)$

Example: Iteration 0

$$x_1 + 2x_2 + x_3 = 3,$$
$$2x_1 + x_2 + x_4 = 3,$$
$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0$$

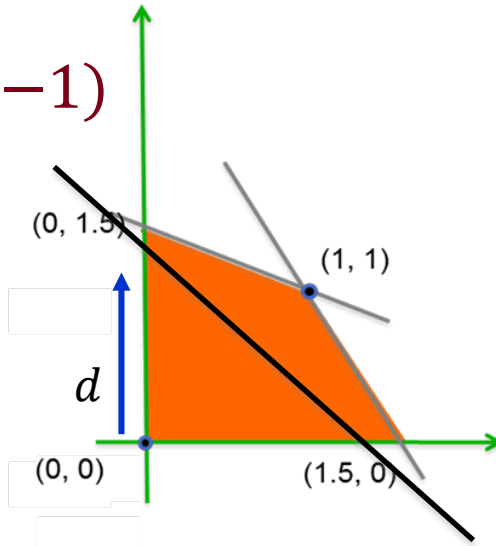


- **Adjacent BFS**
  - $(0, 0, 3, 3) \rightarrow (0, ?, ?, ?)$
- **How far to move?**
  - Stay feasible: $\begin{bmatrix} 0 \\ 0 \\ 3 \\ 3 \end{bmatrix} + \theta \begin{bmatrix} 0 \\ 1 \\ -2 \\ -1 \end{bmatrix} \geq \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$
  - $\Rightarrow \theta \leq 1.5$
  - $(0, 0, 3, 3) + 1.5(0, 1, -2, -1) = (0, 1.5, 0, 1.5)$

Example: Iteration 0

$$
\begin{aligned}
& -\min -x_1 - x_2 \\
& x_1 + 2x_2 + x_3 \quad\quad = 3, \\
& 2x_1 + x_2 \quad\quad + x_4 = 3, \\
& x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0
\end{aligned}
$$

- **Change in cost along direction** $d = (0, 1, -2, -1)$



- $(-1, -1, 0, 0) \begin{bmatrix} 0 \\ 0 \\ 3 \\ 3 \end{bmatrix}$

- $(-1, -1, 0, 0) \begin{bmatrix} 0 \\ 0 \\ 3 \\ 3 \end{bmatrix} + \theta(-1, -1, 0, 0) \begin{bmatrix} 0 \\ 1 \\ -2 \\ -1 \end{bmatrix}$

Reduced Cost

- **Now we consider the objective: $c'x$**
  - $c'x$;
  - $c'(x + \theta d)$;
  - d **is the direction of changing the nonbasic variable with index** j.
- **Reduced cost in direction** j **is defined as**

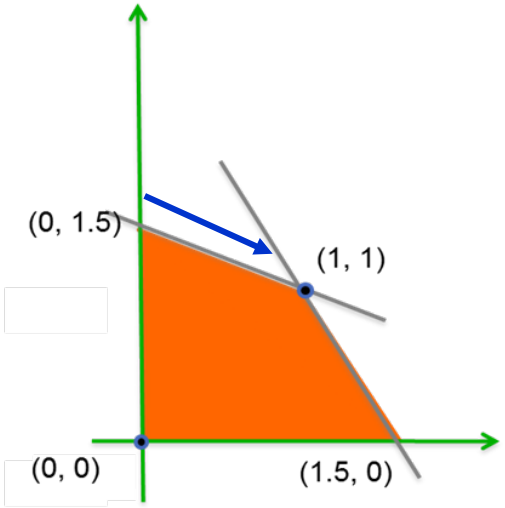
$$\overline{c}_j = \frac{c'(x + \theta d) - c'x}{\theta} = c'd$$

  - **If** $\overline{c}_j < 0$
  - **else**

Example: Iteration 1

$$x_1 + 2x_2 + x_3 = 3,$$
$$2x_1 + x_2 + x_4 = 3,$$
$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0$$

- **Adjacent BFS**
  - $(0, 1.5, 0, 1.5) \rightarrow (?, ?, 0, ?)$
- **Moving direction**
  - $d = (1, d_2, 0, d_4)$



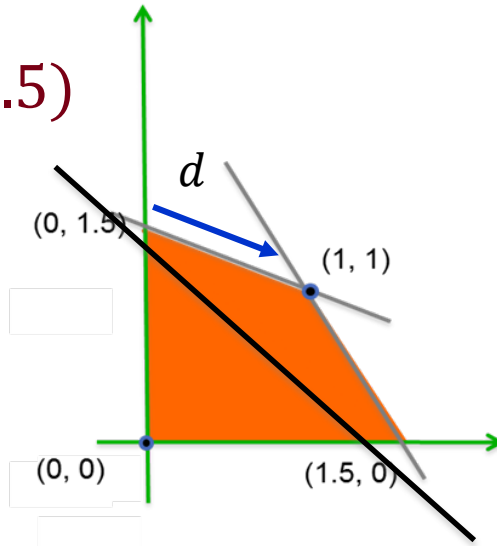
- $$\begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ d_2 \\ 0 \\ d_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 \\ d_2 \\ 0 \\ d_4 \end{bmatrix} = \begin{bmatrix} 1 \\ -0.5 \\ 0 \\ -1.5 \end{bmatrix}$$
  - $(0, 1.5, 0, 1.5) + \theta(1, -0.5, 0, -1.5) \geq \mathbf{0}, \Rightarrow \theta \leq 1$
  - $(0, 1.5, 0, 1.5) + 1(1, -0.5, 0, -1.5) = (1,1,0,0)$

Example: Iteration 1

$$
\begin{aligned}
& -min\ -x_1 - x_2 \\
& x_1 + 2x_2 + x_3 \quad\quad = 3, \\
& 2x_1 + x_2 \quad\quad + x_4 = 3, \\
& x_1 \geq 0, x_2 \geq 0,\ x_3 \geq 0, x_4 \geq 0
\end{aligned}
$$

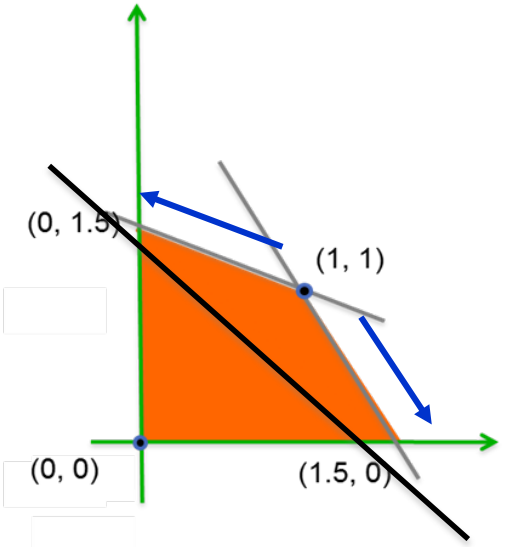
- **Reduced cost in direction:** $d = (1, -0.5, 0, -1.5)$

  - $\bar{c_1} = (-1, -1, 0, 0) \begin{bmatrix} 1 \\ -0.5 \\ 0 \\ -1.5 \end{bmatrix} = -0.5$



Example: Iteration 2

$$-\min -x_1 - x_2$$
$$x_1 + 2x_2 + x_3 = 3,$$
$$2x_1 + x_2 + x_4 = 3,$$
$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0$$



- **BFS:** $(1, 1, 0, 0)$
  - $(1, 1, 0, 0) + \theta(d_1, d_2, 1, 0)$
  - $(1, 1, 0, 0) + \theta(d_1, d_2, 0, 1)$
- **Reduced cost**
  - $\bar{c}_3 = \boldsymbol{c'd} = \frac{1}{3}$
  - $\bar{c}_4 = \boldsymbol{c'd} = \frac{1}{3}$

Optimality Condition and Simplex Method

1. **Find a BFS x;**
2. **Determine whether the current solution is optimal or not: is the reduced cost $\overline{c_j}$ for all the nonbasic variable x_j nonnegative?**
   - **If yes, terminate; else proceed to step 3.**
3. **Pick a nonbasic variable x_j for which the reduced cost is negative and move along this direction, i.e., $\boldsymbol{x} + \theta^* \boldsymbol{d}$**
   - $d_j = 1$;
   - $d_i = 0$, **for** $i \neq j, i \neq B(1), \ldots, B(m)$;
   - $\theta^* = max\{\theta \,|\boldsymbol{x} + \ \theta \boldsymbol{d} \geq \boldsymbol{0}\}$